

CS 4269/5469– Fundamentals of Logic In Computer Science

SEMESTER II, 2025-2026

Home Work-4

- You are welcome to consult any sources you find helpful—books, papers, websites, or even other people. However, you must clearly cite every source you use, and all submitted work must be written in your own words.
 - We strongly discourage the use of AI tools for this homework. The goal of the assignment is to help you engage deeply with the material discussed in class, and relying on AI tends to defeat this purpose. That said, if you do choose to use AI in any way, please acknowledge it and briefly explain how and where it was used. There will be no penalty for doing so; we are simply interested in understanding how these tools are being used.
- You may work either individually or in teams of two. Individual submissions receive a small bonus, but they are expected to require roughly twice the effort of a team submission. If you work in a team, please make sure that the workload is shared fairly.
- For team submissions, all members should be familiar with and have discussed the solutions to every question. This homework is meant to encourage independent thinking and discussion beyond what is covered in lectures.
- Please submit your solutions electronically as a PDF on Canvas. You are encouraged (but not required) to typeset your solutions in $\text{\LaTeX}$. Handwritten submissions are also acceptable.
- The assignment is due on March 19, 2026 (11:59 pm). The grading policy is as follows:
 - Submissions before March 16, 2026 (11:59 pm): 30% extra credit
 - Submissions from March 17, 2026 (12:00 am) to March 17, 2026 (11:59 pm): 15% extra credit
 - Submissions from March 18, 2026 (12:00 am) to March 19, 2026 (11:59 pm): no penalty
 - Submissions from March 20, 2026 (12:00 am) to March 21, 2026 (11:59 pm): 25% penalty
 - Submissions from March 22, 2026 (12:00 am) to March 22, 2026 (11:59 pm): 50% penalty
 - Submissions from March 23, 2026 (12:00 am) onwards: not accepted

Additionally, individual (non-team) submissions receive a 10% bonus applied on top of the above scoring scheme.

First Order Logic (Recap)

Here, we briefly recap the syntax and semantics of FOL¹ for posterity.

Syntax of FOL

Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R})$ be some signature. Here,

- $\mathcal{C}$ is the set of *constant symbols* in Σ ,
- $\mathcal{F}$ is the set of *function symbols* in Σ , and
- $\mathcal{R}$ is the set of *relation symbols* (a.k.a *predicate* or *relational* symbols) in Σ .

Every function symbol $f \in \mathcal{F}$ has an associated *arity*, denoted by $\text{arity}(f) \in \mathbb{N}_{>0}$. Likewise, every relation symbol $R \in \mathcal{R}$ has an associated *arity*, denoted by $\text{arity}(R) \in \mathbb{N}_{>0}$. In order to completely specify the syntax of FOL, we also need a set $\mathcal{V}$ of variables. We will assume that both $\mathcal{V}$ and Σ are countable (i.e., finite or countably infinite)². The syntax of FOL over Σ and $\mathcal{V}$ is given by the following BNF grammar:

$$\begin{aligned} t &:= c \mid x \mid f(t, \dots, t) \\ \varphi &:= R(t, \dots, t) \mid t = t \mid (\neg \varphi) \mid (\varphi \vee \varphi) \mid (\exists x \cdot \varphi) \end{aligned}$$

Here, $c \in \mathcal{C}$, $x \in \mathcal{V}$, $f \in \mathcal{F}$, $R \in \mathcal{R}$ and the number of arguments of f and R match their arities. In the above BNF grammar, ‘ t ’ is the syntactic category of *terms* over Σ and $\mathcal{V}$ and ‘ φ ’ is the syntactic category of *formulae* (or w.f.f.) over Σ and $\mathcal{V}$. The symbol ‘ $\exists$ ’ is the *existential quantifier* and the formula ‘ $\exists x \cdot \varphi$ ’ is read ‘*There exists an x such that φ* ’. We say that φ is an *atomic* formula if it is of the form $\varphi \equiv t_1 = t_2$ or of the form $R(t_1, \dots, t_2)$. We will additionally use the derived operators ‘ $\wedge$ ’, ‘ $\rightarrow$ ’ and the derived *universal* quantifier ‘ $\forall$ ’ which are obtained as follows:

- $\varphi_1 \wedge \varphi_2 \equiv \neg((\neg \varphi_1) \vee (\neg \varphi_2))$
- $\varphi_1 \rightarrow \varphi_2 \equiv (\neg \varphi_1) \vee \varphi_2$
- $\forall x \cdot \varphi \equiv \neg(\exists x \cdot (\neg \varphi))$

Bound and Free Variables

In an FOL formula φ , variables may occur *bound* (to a quantifier) or may occur *free*. For example, in the formula $\varphi_1 = R(\underline{x}) \rightarrow \forall x \cdot T(\underline{x}, f(\underline{x}))$, the first underlined occurrence of x (as part of the sub-formula ‘ $R(x)$ ’) is a *free* occurrence while the later two underlined occurrences are bound (by the quantifier $\forall x$ in the subformula ‘ $\forall x \cdot T(x, f(x))$ ’). The set of bound and free occurrences of

¹FOL stands for ‘First Order Logic’. We will repeatedly use this shorthand throughout the homework and also throughout the course.

²Recall that, when we say that a set is countably infinite, it means that one can assign indices $\{1, 2, 3, \dots\}$ to the elements of the set S so that every element gets a unique index.

a variable $x \in \mathcal{V}$ in a formula can be defined formally in an inductive manner, but is skipped here³.

In general, a variable may occur as both bound and free; for example in the previous formula φ_1 , the variable x occurs as both bound and free. Further, different bound occurrences of the same variable may refer to different quantifiers. For example, in the formula $\varphi_2 = (\forall x \cdot R(x)) \wedge (\exists x \cdot P(x) \wedge (\forall x \cdot Q(x)))$, all the three occurrences of x are bound by different quantifiers. However, bound variables can always be *renamed* to disambiguate this and to ensure that (a) every bound variable is different from every free variable, and (b) two bound occurrences of a variable refer to the same quantifier. For example, the first example can be re-written to an *equivalent* formula $\varphi'_1 = R(x) \rightarrow \forall y \cdot T(y, f(y))$. Similarly, the second formula φ_2 can be rewritten to the equivalent formula $\varphi'_2 = (\forall x \cdot R(x)) \wedge (\exists y \cdot P(y) \wedge (\forall z \cdot Q(z)))$. We use $\text{FVar}(\varphi)$ to denote the set of free variables of a FOL formula φ ; here we go with the notation that a free variable of φ must occur in φ and thus $|\text{FVar}(\varphi)|$ is bounded by the size of the formula φ .

Sentences and Prenex Normal Form

An FOL formula φ is said to be a *sentence* if it has no free variables (i.e., $\text{FVar}(\varphi) = \emptyset$). For example, the formula $\forall x \cdot \forall y \cdot x = y$ is a sentence. Similarly, for a signature in which c and d are constants, the formula $R(c, d) \wedge (S(c) \rightarrow P(d))$ is a sentence.

A formula is said to be *quantifier-free* if it does not have any quantifiers. A formula is said to be in *prenex normal form* if it is of the form $\varphi \equiv Q_1 x_1 \cdot Q_2 x_2 \cdots Q_k x_k \cdot \psi$, where $Q_1, \dots, Q_k \in \{\forall, \exists\}$ are quantifiers, $x_1, \dots, x_k \in \mathcal{V}$ and ψ is quantifier-free (often referred to as the *matrix* of a prenex normal form formula φ). For example, the formula seen before φ_1 is not in prenex normal form as the quantification ' $\forall x$ ' is not outer-most. Any formula not in prenex normal form can be converted to an *equivalent* formula in prenex normal form. For this, we need to first obtain an equivalent formula that does not have any variable reuse and then transform it so that the quantifiers are in the beginning. For example, for φ_1 , the equivalent prenex normal form can be obtained by first converting it to the equivalent formula φ'_1 and then translating it to $\varphi''_1 = \forall y \cdot (R(x) \rightarrow T(y, f(y)))$. Similarly, a prenex normal form equivalent for the formula $\varphi_3 = (\forall x \cdot R(x)) \rightarrow (\forall x \cdot T(x, f(x)))$ is the formula $\varphi'_3 = \exists x \cdot \forall y \cdot (R(x) \rightarrow T(y, f(y)))$. See the detailed Wiki entry on how to convert a given FOL formula into an equivalent FOL formula with the same set of free variables.

Semantics of FOL

The semantics of FOL formulae are given in terms of a set of elements and how each symbol in the signature is interpreted. Fix a signature $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R})$. Formally, a first order Σ -structure (or simply *structure*, when Σ is clear from context) is a pair $\mathcal{A} = (U, \mathcal{I})$, where the *universe* U is a **non-empty** set of elements, and the interpretation $\mathcal{I}$ assigns meaning to all the symbols in Σ :

- For every constant symbol $c \in \mathcal{C}$, the interpretation associates c to some element of the universe, i.e., $\mathcal{I}(c) \in U$.

³Readers are encouraged to look at the succinct Wiki entry on this.

- For every function symbol $f \in \mathcal{F}$, the interpretation assigns a function (or a *mapping*) to the symbol f , i.e., $\mathcal{I}(f) \in [\underbrace{U \times U \cdots U}_{r \text{ times}} \rightarrow U]$, where $r = \text{arity}(f)$. Recall that $[\underbrace{A \times A \cdots A}_{n \text{ times}} \rightarrow A]$ denotes the set of all functions whose domain is the set of n -tuples over A and whose co-domain is A .
- For every relation symbol $R \in \mathcal{R}$, the interpretation assigns a relation to the symbol R , i.e., $\mathcal{I}(R) \subseteq \underbrace{U \times U \cdots U}_{r \text{ times}}$, where $r = \text{arity}(f)$.

Given a universe U , an *assignment* to variables is a mapping $\alpha : \mathcal{V} \rightarrow U$. A structure and a variable assignment completely determine the meaning of a formulae. First, they together assign meanings to each term t , which we denote by $\text{val}_{\mathcal{A},\alpha}(t)$ defined inductively as follows:

$$\begin{aligned} \text{val}_{\mathcal{A},\alpha}(c) &= \mathcal{I}(c) && \text{where } c \in \mathcal{C} \\ \text{val}_{\mathcal{A},\alpha}(x) &= \alpha(x) && \text{where } x \in \mathcal{V} \\ \text{val}_{\mathcal{A},\alpha}(f(t_1, \dots, t_r)) &= \mathcal{I}(f)(\text{val}_{\mathcal{A},\alpha}(t_1), \dots, \text{val}_{\mathcal{A},\alpha}(t_r)) && \text{where } f \in \mathcal{F} \text{ with } \text{arity}(f) = r \end{aligned}$$

Now, we can define the semantics of FOL formulae by inductively defining the relation $\models$ as follows:

$$\begin{aligned} \mathcal{A}, \alpha &\models t_1 = t_2 && \text{iff } \text{val}_{\mathcal{A},\alpha}(t_1) = \text{val}_{\mathcal{A},\alpha}(t_2) \\ \mathcal{A}, \alpha &\models R(t_1, \dots, t_r) && \text{iff } (\text{val}_{\mathcal{A},\alpha}(t_1), \dots, \text{val}_{\mathcal{A},\alpha}(t_r)) \in \mathcal{I}(R) \\ \mathcal{A}, \alpha &\models \neg \varphi && \text{iff } \mathcal{A}, \alpha \not\models \varphi \\ \mathcal{A}, \alpha &\models \varphi_1 \vee \varphi_2 && \text{iff } \mathcal{A}, \alpha \models \varphi_1 \text{ or } \mathcal{A}, \alpha \models \varphi_2 \\ \mathcal{A}, \alpha &\models \exists x \cdot \varphi && \text{iff there is a } u \in U \text{ such that } \mathcal{A}, \alpha[x \mapsto u] \models \varphi \end{aligned}$$

In the above, we use $\alpha[x \mapsto u]$ to denote the new assignment $\alpha' : \mathcal{V} \rightarrow U$ given by:

$$\alpha'(z) = \begin{cases} \alpha(z) & \text{if } z \neq x \\ u & \text{otherwise} \end{cases}$$

Logical Equivalence and Logical Consequence

Two FOL formulae φ_1 and φ_2 are said to be *logically equivalent* if for every structure $\mathcal{A}$ and variable assignment α , we have

$$\mathcal{A}, \alpha \models \varphi_1 \quad \text{iff} \quad \mathcal{A}, \alpha \models \varphi_2$$

Given a set of formulae Γ , structure $\mathcal{A}$ and variable assignment α , we use the notation $\mathcal{A}, \alpha \models \Gamma$ to denote the fact that $\mathcal{A}, \alpha \models \psi$ for every $\psi \in \Gamma$. Given a formula φ and a set of formulae Γ , we say that φ is a *logical consequence* of Γ if for each structure $\mathcal{A}$ and variable assignment α , whenever $\mathcal{A}, \alpha \models \Gamma$, we also have $\mathcal{A}, \alpha \models \varphi$.

Satisfiability, Equisatisfiability and Validity

A formula φ is said to be *satisfiable* if there is a structure $\mathcal{A}$ and a variable assignment α such that $\mathcal{A}, \alpha \models \varphi$; φ is said to be *unsatisfiable* otherwise. A set of formulae Γ is said to be *satisfiable* if there is a structure $\mathcal{A}$ and a variable assignment α such that $\mathcal{A}, \alpha \models \Gamma$. Formulae φ_1 and φ_2 are said to be *equisatisfiable* when either φ_1 and φ_2 are both satisfiable, or when both are unsatisfiable. A formula φ is said to be *valid* (or a *tautology*) if for every structure $\mathcal{A}$ and variable assignment α , we have $\mathcal{A}, \alpha \models \varphi$.

Relevance Lemma

An analogous notion of the Relevance Lemma for FOL is the observation that whenever two variable assignments differ only on values of variables that are not free, then they are equivalent with respect to satisfaction of a formula:

Lemma 1 (Relevance Lemma for FOL). *Let φ be a FOL formula, let $\mathcal{A}$ be a structure and let α_1, α_2 be variable assignments such that for all $x \in \text{FVar}(\varphi)$, $\alpha_1(x) = \alpha_2(x)$, then $\mathcal{A}, \alpha_1 \models \varphi$ iff $\mathcal{A}, \alpha_2 \models \varphi$.*

Question 1. Prove Lemma 1

It follows from the above Relevance Lemma that if φ is a sentence (i.e., $\text{FVar}(\varphi) = \emptyset$), then all variable assignments are equivalent with respect to satisfiability. Thus, in the case of *sentences*, we will simply write $\mathcal{A} \models \varphi$ instead of explicitly specifying a variable assignment.

Herbrand's Theorem for First Order Logic

In our lectures, we saw that the Herbrand's Theorem relates satisfiability of a set of FOL sentences to their propositional satisfiability. Here, we will finish the proof of this theorem. We first recall the definitions of ground terms, ground (atomic) formulae, ground valuations and propositional satisfiability.

Definition 1 (Ground Terms, Atoms and Formulae). A term over a signature Σ is said to be a *ground term* if it does not have any occurrence of a variable in it. We denote the set of all ground terms over Σ by Terms_Σ . A formula is said to be ground if it contains no variables (and thus also no quantifiers). An formula is said to be a *ground atom* if it is both a ground formula and an atomic formula.⁴ We denote by Atoms_Σ and FORM_Σ the set of ground atoms and the set of ground formulae over the signature Σ . We remark that if $\mathcal{C} \neq \emptyset$, then Terms_Σ , Atoms_Σ and FORM_Σ are non-empty. Further, if additionally $\mathcal{R} \neq \emptyset$, then the set of ground atoms and ground formulae that do not use the equality symbol '=' are also non-empty.

Definition 2 (Ground Instantiation). Let $\varphi \equiv \forall x_1 \forall x_2 \dots \forall x_n \psi$ be a universally quantified FOL sentence over Σ and $\mathcal{V}$ in prenex normal form (and thus the matrix ψ is quantifier-free). A ground instantiation of φ is a formula ϕ obtained by replacing, for every $1 \leq i \leq n$, each occurrence of x_i in ψ by some ground term t_i . That is, there is a mapping $T : \mathcal{V} \rightarrow \text{Terms}_\Sigma$ such that $\phi \equiv \psi[x_1 \rightarrow T(x_1)] \dots [x_n \rightarrow T(x_n)]$. We use $\text{Ground}(\varphi)$ to denote the set $\{\psi[x_1 \rightarrow T(x_1)] \dots [x_n \rightarrow T(x_n)] \mid T : \mathcal{V} \rightarrow \text{Terms}_\Sigma\}$ of all ground instantiations of φ . For a set Γ of universally quantified FOL sentences over Σ and $\mathcal{V}$, we denote $\text{Ground}(\Gamma) = \bigcup_{\varphi \in \Gamma} \text{Ground}(\varphi)$.

Definition 3 (Propositional Satisfiability). A *ground atomic valuation* over Σ is a mapping $\tau : \text{Atoms}_\Sigma \rightarrow \{\text{true}, \text{false}\}$. Such a valuation can be extended to ground formulae in the natural manner, given by the relation $\models_{\text{Gr}}$:

$\tau \models_{\text{Gr}} \psi$	iff	$\tau(\psi) = \text{true}$	if $\psi \in \text{Atoms}_\Sigma$
$\tau \models_{\text{Gr}} (\neg \varphi)$	iff	$\tau \not\models_{\text{Gr}} \varphi$	if φ is ground
$\tau \models_{\text{Gr}} (\varphi_1 \vee \varphi_2)$	iff	$\tau \models_{\text{Gr}} \varphi_1$ or $\tau \models_{\text{Gr}} \varphi_2$	if φ_1, φ_2 are ground

⁴Observe that ground atom and ground formulae are also *sentences* as they do not have any free variables.

A set of ground FOL formulae Γ is said to be *propositionally satisfiable* if there is a ground atomic valuation τ such that $\tau \models_{\text{Gr}} \varphi$ for every $\varphi \in \Gamma$.

We are now ready to state (a simpler version of) the Herbrand's theorem. We will only talk about FOL formulae that are universally quantified sentences that additionally do not have the symbol '='. We will refer to them as *formulae without equality*.

Theorem 1 (Herbrand's Theorem without Equality). Assume that the signature Σ has at least one constant (i.e., $\mathcal{C} \neq \emptyset$). Let Γ be a set of universally quantified FOL sentences without equality in prenex normal form. Γ is satisfiable iff $\text{Ground}(\Gamma)$ is propositionally satisfiable.

In the following we will fix the signature Σ and assume that it has at least one constant. We will also assume the set Γ which is a set of universally quantified FOL sentences without equality in prenex normal form. Let us first get the easier direction out of the way.

Lemma 2. *If Γ is satisfiable then $\text{Ground}(\Gamma)$ is propositionally satisfiable.*

Question 2. Prove Lemma 2 **Hint:** Consider a structure $\mathcal{A}$ such that $\mathcal{A} \models \Gamma$. Define a ground atomic valuation $\tau_{\mathcal{A}}$ that assigns true or false to some ground atom ϕ based on whether the ground atom evaluates to true or false in $\mathcal{A}$. Now argue, that for every ground formula $\psi \in \text{Ground}(\Gamma)$, we have $\tau_{\mathcal{A}} \models_{\text{Gr}} \psi$; remember that ψ may not, in general, be atomic and this is where structural induction will help.

Let us now progress towards the proof of Theorem 1 and prove the opposite direction:

Lemma 3. *If $\text{Ground}(\Gamma)$ is propositionally satisfiable then Γ is satisfiable.*

The proof of the above lemma relies on the observation that even though the different structures that witness the satisfiability of a set Γ (of universally quantified FOL sentences) may be arbitrarily complex, there is a special simple structure, called the Herbrand structure that will also witness the satisfiability of Γ . We describe this special structure next.

Definition 4 (Herbrand Structure). Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R})$ be a FOL signature with $\mathcal{C} \neq \emptyset$. The Σ -Herbrand structure is a tuple $\mathcal{A}_{\Sigma} = (U_{\Sigma}, \mathcal{I}_{\Sigma})$ given as follows:

- $U_{\Sigma} = \text{Terms}_{\Sigma}$ is the unique set of ground terms over Σ . Observe that this is a non-empty set. This is often called the *Herbrand universe*.
- $\mathcal{I}_{\Sigma}$ assigns the unique 'natural' interpretation to the constants and function symbols in Σ (also called the *Herbrand interpretation*). That is,
 - For every constant symbol $c \in \mathcal{C}$, $\mathcal{I}_{\Sigma}(c) = c$ (and thus $\mathcal{I}_{\Sigma}(c) \in U_{\Sigma}$).
 - For every function symbol $f \in \mathcal{F}$ having arity $\text{arity}(f) = r$, and for every r -tuple $(t_1, t_2, \dots, t_r) \in U_{\Sigma}$, $\mathcal{I}_{\Sigma}(f)(t_1, \dots, t_r) = f(t_1, \dots, t_r)$; again observe that $\mathcal{I}_{\Sigma}(f) \in [U_{\Sigma} \times \dots \times U_{\Sigma} \rightarrow U_{\Sigma}]$.

Observe that $\mathcal{I}_{\Sigma}$ does not restrict what interpretations must be given to the relation symbols, and any meaning for symbols in $\mathcal{R}$ can be picked as part of $\mathcal{I}_{\Sigma}$.

Question 3. Prove Lemma 3 **Hint:** Start with a ground atomic valuation τ that witnesses the propositional satisfiability of $\text{Ground}(\Gamma)$. Construct a Herbrand Structure $\mathcal{A}_\Sigma^\tau$ by assigning interpretations to the relation symbols according to τ . Show that the resulting structure makes every formula in Γ true.

Implications of Herbrand's Theorem

The proof of Herbrand's Theorem, we saw so far, highlights some very interesting *model-theoretic* observations. We outline them here and ask you to prove them.

First, the proof illustrates that whenever there is a model $\mathcal{A}$ for a set Γ of universally quantified FOL sentences without equality⁵, there is always a model which is not *too large*. This is often called the **Downward Löwenheim-Skolem Theorem**⁶.

Theorem 2 (Downward Löwenheim-Skolem Theorem). Let Σ be a countable structure and let Γ be a set of (countable or uncountable) universally quantified FOL sentences without equality. If Γ is satisfiable, then there is a countable structure $\mathcal{A}$ such that $\mathcal{A} \models \Gamma$.

Question 4. Prove Theorem 2. **Hint:** Are Herbrand structures countable?

Next, let us highlight another subtlety in the proof. The proof crucially uses the technique of ground instantiations to reason about satisfiability. The core idea here is to construct an *equisatisfiable* set of formulae. Care must be taken to avoid incorrectly concluding that ground instantiations lead to a model-theoretically *equivalent* set of formulae. Let us illustrate this point in detail. First, let us make the following observation.

Theorem 3. Γ is satisfiable iff $\text{Ground}(\Gamma)$ is satisfiable.

You are not asked to prove Theorem 3 (SIGH!), but I expect that you should be able to prove it if you have followed the proof of Herbrand's Theorem. Notice that the above statement talks about *equisatisfiability*. It does not talk about *equivalence* — we say two sets of sentences Γ_1 and Γ_2 are equivalent if for every FOL structure $\mathcal{A}$, $\mathcal{A} \models \Gamma_1$ iff $\mathcal{A} \models \Gamma_2$. In fact, such a statement would be incorrect in general. Your task is to come up with an example that demonstrates this.

Question 5. Describe a signature Σ , a set Γ of universally quantified FOL sentences without equality over Σ and a Σ -structure $\mathcal{A}$ such that $\mathcal{A} \models \text{Ground}(\Gamma)$ but $\mathcal{A} \not\models \Gamma$. **Hint:** Try to come up with a small Σ , a small Γ and a small $\mathcal{A}$ to make your life easier.

⁵Later, we will be able to conclude that the restriction of universally quantified formulae and the restriction of 'no equality' can be lifted.

⁶A dual result, called the **Upward Löwenheim-Skolem Theorem** states that if there is a model for a set Γ , then, in fact, there is a model of arbitrary cardinalities. That is, if Γ is satisfiable, then one can come up with models whose cardinalities are $|\mathbb{N}|$, $|\mathcal{P}(\mathbb{N})|$, $|\mathcal{P}(\mathcal{P}(\mathbb{N}))|$, ... and so on. In other words, an FOL formula, or even a set of FOL formulae, cannot distinguish between countable and uncountable models (or models of higher cardinality).

The Case of Equality

So far, we have proved Herbrand's theorem for the case when of formulae that do not contain any equality. This simplifies our proof, but more importantly, as we will show now, that the case when we have equality can be translated to the case without equality. First, let us try to understand the subtlety in reasoning with equality. Recall that, Theorem 1 reduces satisfiability to propositional satisfiability of ground instantiations. In other words, a ground atomic valuation that makes all of $\text{Ground}(\Gamma)$ propositionally satisfiable guarantees the existence of a FO structure that satisfies the entire Γ . In the presence of equality, this is not true. The next question asks you to show that this is a problem even when Γ is already ground and also singleton!

Question 6. Let $\Sigma = (\{c, d, e\}, \{f\}, \{R\})$ be a signature with three constants (c, d and e), a unary function symbol f and a unary relation symbol R (i.e., $\text{arity}(f) = \text{arity}(R) = 1$). Let

$$\begin{aligned}\psi_1 &\equiv \neg(c = c) & \psi_2 &\equiv (c = d) \wedge \neg(d = c) \\ \psi_3 &\equiv (c = d) \wedge (d = e) \wedge \neg(c = e) & \psi_4 &\equiv (c = d) \wedge \neg(f(c) = f(d)) \\ \psi_5 &\equiv (c = d) \wedge R(c) \wedge \neg(R(d))\end{aligned}$$

Let $\Gamma_i = \{\psi_i\}$ (for $i \in \{1, 2, 3, 4, 5\}$). Show that, for each i , $\text{Ground}(\Gamma_i)$ is propositionally satisfiable but Γ_i is not satisfiable.

The above example illustrates an important observation — the equality symbol '=' is special; it cannot be treated as an arbitrary binary relation symbol in the signature. Indeed, as the example above illustrates, not every ground atomic valuation makes sense in the first order sense. However, as far as checking satisfiability is concerned, we can navigate this issue by essentially restricting the space of ground atomic valuations to those that respect certain properties. These properties are often termed as the *equality axioms*:

$$\begin{aligned}\text{E1: } & \forall x \cdot x = x & & \text{(Reflexivity)} \\ \text{E2: } & \forall x, y \cdot (x = y) \rightarrow (y = x) & & \text{(Symmetry)} \\ \text{E3: } & \forall x, y, z \cdot ((x = y) \wedge (y = z)) \rightarrow (x = z) & & \text{(Transitivity)} \\ \text{E4: } & \forall x_1, \dots, x_r, y_1, \dots, y_r \cdot \bigwedge_{i=1}^r (x_i = y_i) \rightarrow f(x_1, \dots, x_r) = f(y_1, \dots, y_r) & & \text{(Func. Congruence)} \\ \text{E5: } & \forall x_1, \dots, x_r, y_1, \dots, y_r \cdot \bigwedge_{i=1}^r (x_i = y_i) \rightarrow (R(x_1, \dots, x_r) \leftrightarrow R(y_1, \dots, y_r)) & & \text{(Rel. Congruence)}\end{aligned}$$

In the above, we have one axiom (E4) for every function symbol f in Σ and one axiom (E5) for every function symbol R in Σ ; the number of quantifiers in each of these axioms is appropriately selected depending upon the arity of these symbols.

Observe that every structure $\mathcal{A}$ implicitly satisfies all of these axioms. On the other hand, ground atomic valuations may not reflect these axioms accurately when we naively treat equality as an arbitrary binary relation (as in Question 5). However, we do want to stick to the same (or atleast similar) reasoning framework — reduce satisfiability to propositional satisfiability, in the spirit of the simpler Herbrand's theorem we have seen before. This framework is simple, intuitive and reduces to the much more familiar propositional reasoning! We will circumvent this by treating the '=' symbol as a usual binary relation symbol but instead change the original set of formulae Γ so that only meaningful ground atomic valuations are allowed. Let us formalize this next.

Proxy binary relation for equality

We will first augment the original signature $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R})$ (over which Γ is written) by adding a fresh binary relation symbol 'EQ' (i.e., $\text{arity}(f) = 2$ and $\text{EQ} \notin \mathcal{R}$). Let $\Sigma^* = (\mathcal{C}, \mathcal{F}, \mathcal{R} \uplus \{\text{EQ}\})$.

Now, we will rewrite formulae that use the '=' symbol by replacing $=$ with EQ. Formally, given a FOL formula φ , we define $\text{eq}(\varphi)$ to be the formula where every atomic formula of the form $t_1 = t_2$ is replaced by the formula $\text{EQ}(t_1, t_2)$; the definition of $\text{eq}(\varphi)$ can be stated in an inductive manner in a straightforward manner and we skip it. As an example, for the formula $\phi \equiv (c = d) \wedge \forall x, y. f(x, y) = f(y, x)$, we have $\text{eq}(\phi) \equiv \text{EQ}(c, d) \wedge \forall x, y. (\text{EQ}(f(x, y), f(y, x)))$. Observe that if φ was over signature Σ , then $\text{eq}(\varphi)$ will be over the signature Σ^* as defined in the previous paragraph. Further, the quantification structure of φ and $\text{eq}(\varphi)$ is identical; thus φ is quantifier-free, universally quantified or in prenex normal form iff $\text{eq}(\varphi)$ is respectively quantifier-free, universally quantified, in prenex normal form. For a set Γ of FOL formulae, we use the notation $\text{eq}(\Gamma) = \{\text{eq}(\varphi) \mid \varphi \in \Gamma\}$.

Adding the axioms for Equality

As discussed multiple times now, simply substituting $=$ with a new binary relation symbol is not going to be adequate (see for yourself that $\text{eq}(\text{Ground}(\Gamma_i))$ in Question 5, is also propositionally satisfiable). To take care of this, we consider the axioms of equality that we saw before, this time for the new symbol EQ instead of $=$. In this light, let us define the following five sets of FOL formulae:

$$\mathcal{E}_1 = \{\forall x. \text{EQ}(x, x)\}$$

$$\mathcal{E}_2 = \{\forall x, y. \text{EQ}(x, y) \rightarrow \text{EQ}(y, x)\}$$

$$\mathcal{E}_3 = \{\forall x, y, z. (\text{EQ}(x, y) \wedge \text{EQ}(y, z)) \rightarrow \text{EQ}(x, z)\}$$

$$\mathcal{E}_4 = \{\forall x_1, \dots, x_r, y_1, \dots, y_r. (\bigwedge_{i=1}^r \text{EQ}(x_i, y_i)) \rightarrow \text{EQ}(f(x_1, \dots, x_r), f(y_1, \dots, y_r)) \mid f \in \mathcal{F}, \text{arity}(f) = r\}$$

$$\mathcal{E}_5 = \{\forall x_1, \dots, x_r, y_1, \dots, y_r. (\bigwedge_{i=1}^r \text{EQ}(x_i, y_i)) \rightarrow (R(x_1, \dots, x_r) \leftrightarrow R(y_1, \dots, y_r)) \mid R \in \mathcal{R}, \text{arity}(f) = r\}$$

Let $\mathcal{E} = \bigcup_{i=1}^5 \mathcal{E}_i$. Observe that if the signature Σ is countable, then $\mathcal{E}$ is also a countable set.

Given a set Γ of FOL formulae over Σ , we use $\Gamma^* = \text{eq}(\Gamma) \cup \mathcal{E}$. Observe that if Γ was a set of universally quantified FOL sentences (with or without equality), then so is Γ^* , and additionally, Γ^* does not use the equality symbol. Voila!

We can now state the final ingredient that will help us get to the more general Herbrand's Theorem.

Theorem 4. Γ is satisfiable iff Γ^* is satisfiable.

Let us first prove the easier direction

Lemma 4. *If Γ is satisfiable then Γ^* is satisfiable.*

Question 7. Prove Lemma 4. **Hint:** Consider a structure $\mathcal{A}$ such that $\mathcal{A} \models \Gamma$. Can we construct a structure $\mathcal{A}^*$ by augmenting $\mathcal{A}$ with the interpretation of the new symbol EQ?

What interpretation will you assign to EQ in this new structure? You will have to inductively prove that $\mathcal{A}^* \models \varphi$ for every $\varphi \in \Gamma^*$.

Let us now turn to the more involved direction.

Lemma 5. *If Γ^* is satisfiable then Γ is satisfiable.*

We will look at a proof that builds a structure $\mathcal{A}$ for Γ by *shrinking* a structure $\mathcal{A}^*$ that witnesses the satisfiability of Γ^* . To formalize this, let us look at some definitions.

Definition 5 (Equivalence Relation and Classes). Let U be a set and let $R \subseteq U \times U$ be a binary relation over U . R is said to be an *equivalence* relation if it satisfies the following properties:

- R is reflexive. That is, for every element $u \in U$, we have $(u, u) \in R$.
- R is symmetric. That is, for any two elements $u, v \in U$, if $(u, v) \in R$, then $(v, u) \in R$.
- R is transitive. That is, for any three elements $u, v, w \in U$, if $(u, v) \in R$ and if $(v, w) \in R$, then $(u, w) \in R$.

An equivalence relation partitions the set into equivalence *classes*. For an element $u \in U$, we denote the equivalence class of u (induced by R) by $[u]_R = \{v \mid (u, v) \in R\}$. The *quotient* set of U with respect to the equivalence relation R , denoted U/R is the set of equivalence classes induced by R . That is, $U/R = \{[u]_R \mid u \in U\}$.

As an example, given a directed graph $G = (V, E)$, the relation $\text{Reach2} = \{(u, v) \mid u \text{ is reachable from } v \text{ and } v \text{ is reachable from } u\}$ over V is an equivalence relation. In this case, the equivalence class of a vertex v with respect to Reach2 is the set of vertices that can reach v and are reachable from v , also known as the maximal strongly connected component (SCC) that v belongs to. The quotient of V with respect to Reach2 is the set of maximal SCCs of G .

Definition 6 (Congruence Relation). Let U be a set and let $E \subseteq U \times U$ be a binary relation over U . Let $\mathcal{F}$ be a set of functions and $\mathcal{R}$ be a set of relations over U ; each function and relation has an associated arity. E is said to be a *congruence* relation with respect to $\mathcal{F}$ and $\mathcal{R}$ if it satisfies the following properties:

- E is an equivalence relation.
- E preserves equivalence with respect to every function. That is, for every function $f \in \mathcal{F}$ (of arity r), and for any $2r$ elements $u_1, u_2, \dots, u_r, v_1, \dots, v_r \in U$, if $E(u_1, v_1), E(u_2, v_2), \dots$, and $E(u_r, v_r)$, then $E(f(u_1, \dots, u_r), f(v_1, \dots, v_r))$.
- E is consistent with respect to every relation. That is, for every function $R \in \mathcal{R}$ (of arity r), and for any $2r$ elements $u_1, u_2, \dots, u_r, v_1, \dots, v_r \in U$, if $E(u_1, v_1), E(u_2, v_2), \dots$, and $E(u_r, v_r)$, then either both $(u_1, \dots, u_r) \in R$ and $(v_1, \dots, v_r) \in R$ or both $(u_1, \dots, u_r) \notin R$ and $(v_1, \dots, v_r) \notin R$.

Observe that, for a set U , the identity relation $\text{id}_U = \{(u, u) \mid u \in U\}$ is an equivalence and a congruence relation with respect to every function and relation over U .

Let us get back to proving Lemma 5.

Question 8. Let $\mathcal{A} = (U, \mathcal{I})$ be a structure for the signature $\Sigma^* = (\mathcal{C}, \mathcal{F}, \mathcal{R} \uplus \{\text{EQ}\})$ such that $\mathcal{A} \models \mathcal{E}$. Show that $\mathcal{I}(\text{EQ})$ is a congruence relation with respect to every function symbol $f \in \mathcal{F}$ and relation symbol $R \in \mathcal{R}$.

Question 9. Prove Lemma 5. **Hint:** Let $\mathcal{A}^* = (U^*, \mathcal{I}^*)$ be a structure for Γ^* . Using the Question 7, first show that $\mathcal{A}^*$ interprets EQ as a congruence. Next, construct a new structure $\mathcal{A} = (U, \mathcal{I})$ whose universe U is the quotient set $U^*/\mathcal{I}^*(\text{EQ})$. How would you define the interpretation $\mathcal{I}$? Now establish (using structural induction) that $\mathcal{A} \models \varphi$ for every $\varphi \in \Gamma$.

Putting it all together

Theorem 5. Assume that the signature Σ has at least one constant (i.e., $\mathcal{C} \neq \emptyset$). Let Γ be a set of universally quantified FOL sentences in prenex normal form⁷. Γ is satisfiable iff $\text{Ground}(\Gamma^*)$ is propositionally satisfiable.

Question 10. Prove Theorem 5 by putting together all the pieces so far.

⁷No restriction on equality this time